

Study Abroad Mentor Platform — Implementation Plan

Oct 2, 2026 · @Husnain Aslam

1. Summary

Four part-time students can open a small private beta on **Mon 14 Dec 2026** and launch publicly for Pakistan → Germany and Pakistan → UK on **Mon 22 Feb 2027**. That works only if the first version stays small and the team follows a fixed weekly rhythm.

The honest maths. Each person gives 10–15 hours a week. After meetings, reviews, testing and learning, the team has about **16 hours of feature coding a week**, roughly one full-time developer. Claude speeds coding up by an estimated 1.4×, but it does not remove reviews, testing or vendor paperwork. So the plan cuts scope rather than move the season.

Why these dates. Demand for both corridors peaks from about February to May 2027:

- UCAS's main undergraduate deadline is 13 Jan 2027 ([UCAS](#)).
- German uni-assist submissions mostly run April to July, with a general 15 July deadline ([Uni Passau](#)).
- The earliest you can apply for a UK Student visa is 6 months before the course starts ([GOV.UK](#)), so September-start students begin around March.

What you build:

- **Private beta:** verified mentors, booking requests, Safepay checkout, held payments with a tamper-proof ledger, LiveKit video sessions, the contact filter, reminder emails and an in-app admin page with 2FA. Some steps are done by hand, but only in safe ways.
- **Public launch adds:** automatic payment confirmation (webhook), a mentor calendar with instant booking, attendance evidence, parent-linked accounts for under-18s, a free intro call with refund rules, checklist pages and booking messages.
- **After launch:** the AI features, real-time chat, Urdu interface and Ask-a-Senior.

How the team works:

- **Roles:** you (Backend) are also the product owner. The AI Engineer is the second code owner for money and security code. The R&D Specialist runs research, mentor recruitment, usability tests and QA coordination.

- **Rhythm:** a short async standup every working day, a mid-week check on **Wednesday at 22:00**, a weekly sync on **Saturday at 11:00** and a test bash on **Sunday at 20:00** (all PKT).
- **AI tools:** Claude Code does MAJOR tasks (at most 3 sessions per person per week). Antigravity does MINOR ones. **No real user data ever goes into any AI tool.**
- **Testing:** about 20% of everyone's time. Usability, performance, reliability and fault-tolerance tests each have dates and pass rules.

How this plan was made. Four specialist agents (delivery, QA, AI workflow, architecture) each drafted a part. They then raised 40 challenges against each other's drafts, 8 of them blockers. The authors answered and revised. Every blocker changed the plan, for example "show, don't send" identity checks and no Supabase dashboard access for non-coders.

Working detail lives in three tabs:

- **Day-by-day plan** : what each person does each day in weeks 1–6.
- **Testing playbook** : test cases, scripts, the fault table and gate checklists.
- **AI coding playbook** : prompts, config files and safe settings.

Decisions only you can make (by the date shown):

Decision	Options	Needed by
Company or sole proprietor	Decide after the lawyer consult	Sat 17 Oct
Refund details D1–D6	Section 10 of the Testing playbook tab	Sat 31 Oct
Supabase plan for the waitlist	Free project from week 1, which pauses after a week without activity (Supabase), or Pro at \$25/month from October	Mon 5 Oct
Beta without real money	If Safepay live keys or the lawyer's OK miss Fri 11 Dec, run free or vouchered sessions and lower the launch gate's "30 paid sessions"	Fri 11 Dec
Under-18 UK applicants	They wait until guardian accounts ship at launch, missing the January UCAS rush	Sat 14 Nov
R&D overload (~98% before the beta)	You take over mentor screening calls when needed, or recruit a fifth volunteer	Sat 31 Oct

Decision	Options	Needed by
Vercel seats	Vercel's free Hobby plan is non-commercial only, so production needs Pro (\$20/month per seat). Ask Vercel whether one CI deploy token covers the team or every committer needs a seat	Before Sat 28 Nov

2. MVP scope

The first version has 15 "Must" items for the private beta (~103.5 h) and 7 ranked "Should" items for public launch (~60 h). The architect estimated ~545 h for the full concept, so everything else waits. Hours are estimates **with AI help**, including each item's own tests.

MoSCoW sorts work into **Must** (needed for the private beta), **Should** (needed for public launch), **Could** (soon after) and **Won't** (not now). Both corridors share one codebase. "Corridor" is just a field on mentors, gigs and fact cards, plus different mentor-eligibility rules:

- **UK:** paid mentors must hold a Graduate visa, settled status or citizenship, never a Student visa.
- **Germany:** student mentors need written approval from their foreigners' office (Ausländerbehörde).

Must: private beta, Mon 14 Dec 2026

ID	Item	FE	BE	AI	Total h
M1	Foundation: monorepo, CI, staging + production Supabase, sign-in (email + Google), 3 roles, Sentry. RLS (database rules for who sees which rows): you write the access matrix and the policies; the AI Engineer writes the tests separately	1	6	5	12
M2	Admin 2FA (authenticator-app code) on admin pages and routes	0	0	2	2
M3	Translation-ready shell: English only for now, but built so Urdu is translation work later, not a rewrite	1.5	0	0	1.5
M4	Landing page, waitlist, legal pages ("admission never guaranteed", "no visa advice"), 18+ gate	4	0	0	4

ID	Item	FE	BE	AI	Total h
M5	Mentor application + document upload to a private bucket; files deleted after verification. Fake documents only on staging	6	2	1	9
M6	Mentor directory + filters (corridor, field, price, female mentor) + profile with 2 gigs	7	1	0	8
M7	Booking request: student proposes 3 times, mentor accepts one; both time zones shown; no double booking; session at least 36 h after acceptance	4	7	0	11
M8	Payment v0: Safepay checkout, link expires 24 h before the session; admin "mark paid" needs a unique Safepay reference + exact amount; append-only ledger that cannot be edited	1	8.5	0	9.5
M9	Session: LiveKit video room, audio-only button, "technical problem → free reschedule", Confirm with a 1-5 rating, Dispute, auto-confirm after 72 h	3.5	1.5	3	8
M10	Contact filter: one database function + one test list of tricks (Urdu digits, "O" for zero, spaced digits, +49/+44 numbers); hard block only for clear cases	0.5	0	5	5.5
M11	"My bookings" page + emails + automatic reminders 24 h and 1 h before	0	0	7.5	7.5
M12	In-app admin with 2FA: verification queue for R&D; mark paid / refund / payout for you and the AI Engineer (payout sheet never shows full IBANs)	1	2	5	8
M13	First-party metrics: events table + funnel view (no cookies)	0	0	2	2
M14	Mobile polish, empty/error states, basic accessibility	1.5	0	0	1.5
M15	Beta hardening: usability fixes, production set-up, separate job worker, k6 smoke test, nightly backups, restore drill #1	4	4	6	14
	Totals	35	32	36.5	103.5

Capacity before the beta is ~128 coding hours, so the load is about **81%**. The remaining ~20% covers AI-limit waits, reviews and surprises.

Done by hand in the beta (safe versions only)

Job	How, safely	Automated by
Mentor verification	Until Sat 28 Nov, "show, don't send": the mentor shows the document on a video call; R&D logs metadata only (type, valid-until, who checked). From Mon 30 Nov mentors upload into the private bucket	Stays a human check
Matching students to mentors	A short form with no phone or grades; R&D replies with 3 profile links within 24 h	Smart onboarding (after launch)
Payment confirmation	You "mark paid" twice a day with the Safepay reference + amount. Without Safepay live keys, sessions are free or vouchered. Never bank or Raast transfers	Webhook live Mon 25 Jan
Refunds and payouts	Refund in the Safepay dashboard, then record it; weekly payout on Tuesday from the 2FA admin page	Refund rules at launch
Under-18 students	Not allowed in the beta	Guardian accounts at launch
Support	WhatsApp Business app on a dedicated SIM, inbound only, in Urdu; logs use booking codes only	–

Should: public launch, Mon 22 Feb 2027 (ranked; drop from the bottom if late)

Rank	Item	Hours
L1	Payment webhook + reconciliation (never dropped). Signature check written by hand; runs in "shadow mode" from Sat 19 Dec, which records what it would do but changes nothing; goes live Mon 25 Jan	18

Rank	Item	Hours
L2	Mentor availability calendar + instant booking with slot holds	9
L3	Attendance evidence (LiveKit events + our own heartbeat); unclear cases go to an admin, never an automatic refund	8
L4	Parent-linked accounts for under-18s + read-only parent view	9
L5	Free 15-min intro call + refund rules (proposed by the system, approved with one admin click)	6
L6	Germany/UK checklist and roadmap pages, built from verified fact cards, with "last verified" dates and "Report wrong info"	6
L7	Booking message thread (refresh-based, contact-filtered)	4

Could, soon after launch (S11–S14, March–April 2027)

- AI personalised checklist and roadmap (retrieval-based answers with official sources, guardrails, cost caps)
- Smart onboarding form
- Real-time chat with AI moderation
- Ask-a-Senior
- Admin v2
- SEO pages
- WhatsApp reminders through Meta's Cloud API
- The Urdu interface

Later, after Gate G2 (100 paid sessions)

- Communities
- Full AI university and scholarship finder
- My Journey tracker
- Admits & Rejects board

- Paid webinars
- Mentor Pro and boosted listings
- Arrival Kit
- AI pre-check and interview practice
- German Speaking Buddy
- Converting admitted students into mentors (a manual invite email until then)

Won't, not in this plan

Cut	Why
AI professor finder	A niche need, ~30 h of integration, not linked to booking revenue
Accommodation finder	Needs signed property partners in two countries
University portal	B2B sales take months and need student traffic first
AI session recordings	Consent, voice-data storage and per-minute cost are too risky for an MVP
Instalments	They add ledger risk to small tickets
Native apps	The installable website (PWA) covers mobile
International payouts	Need a foreign company. Mentors need a Pakistani bank account for now
Stored wallet balances	Need a State Bank of Pakistan e-money licence
India	Comes after G2

3. Team roles and ownership

Every area has one owner and one backup, so nothing stops if one person is ill or in exams. You hold two roles, Backend and product owner, so you get 14 h a week planned instead of 12.5.

Where each person's week goes (normal week, estimate)

Hours per week	Frontend + UI/UX	Backend (you, PO)	AI Engineer	R&D Specialist
Committed	12.5	14	12.5	12.5
Meetings (sync, sprint events, standups)	2	2	2	2
Product-owner work	–	2	–	–
Code reviews	1	1.5	1.5	–
Learning, set-up, waiting for AI limits	1.5	1.5	1	1
Testing	2.5	2	2.5	3
Building time	5.5	5	5.5	6.5 (non-code)

What each role does

Role	Builds	Tests	Non-coding duties
Frontend + UI/UX	Figma designs (high-fidelity only for tested flows), all screens, the translation-ready shell, installable PWA, accessibility, mobile speed	Component tests (Vitest), browser journeys (Playwright + axe), checks on 3 Android phones + 1 iPhone; co-moderates usability rounds	4 student interviews; landing-page copy; social-media visuals
Backend (you)	Data model, access matrix written by hand , RLS policies, booking logic, Safepay + ledger + payouts, webhook signature check (by hand), production set-up	API and money tests (pytest + Hypothesis), webhook fault tests, access-control tests	Product-owner duties (below); signs for lawyer, SECP, bank and Safepay; runs payouts and daily reconciliation; at least 5 student interviews

Role	Builds	Tests	Non-coding duties
AI Engineer	Before launch a full-stack generalist: CI, Sentry, 2FA, contact filter, My bookings, emails and reminders, job worker, LiveKit, in-app admin, metrics, backups. After the beta: webhook jobs, calendar slots, attendance, guardian API. From March: the AI features	RLS tests from your matrix (in a separate session from the policies), filter test list, k6 load tests, restore drills, nightly database checks	Second code owner (reviews your money/security PRs); trained backup for payouts; emergency production access; 4 student interviews
R&D Specialist	Germany/UK research; fact cards (drafted with Claude chat, every fact checked on the official page); mentor recruitment and video checks; legal and payment paperwork	Runs the Sunday test bash; writes test cases for key flows; exploratory testing as a "student"; first-pass bug triage; signs the release checklist	12+ student interviews (coded notes); community seeding; webinars; Urdu support rota lead

Owner and backup for each area

Area	Owner	Backup
Backlog, priorities, scope	You (PO)	R&D
UI/UX design and frontend code	Frontend	AI Engineer
Backend API, schema, RLS policies	You	AI Engineer (writes the RLS tests)
Payments, webhook, ledger, payouts	You	AI Engineer
Infrastructure, deploys, secrets	You	AI Engineer
Backups and restore drills	AI Engineer	You

Area	Owner	Backup
Contact filter, notifications, jobs, metrics	AI Engineer	You
AI features and LLM cost limits (from March)	AI Engineer	You
Fact cards and their freshness	R&D	AI Engineer
Mentor recruitment and verification	R&D	You
Usability tests, QA coordination, support	R&D	Frontend

Bus-factor rule. The "bus factor" is how many people you could lose before work stops. Each backup reviews at least one PR per sprint in the area they back up, and pairs with the owner for 1 h before every release there.

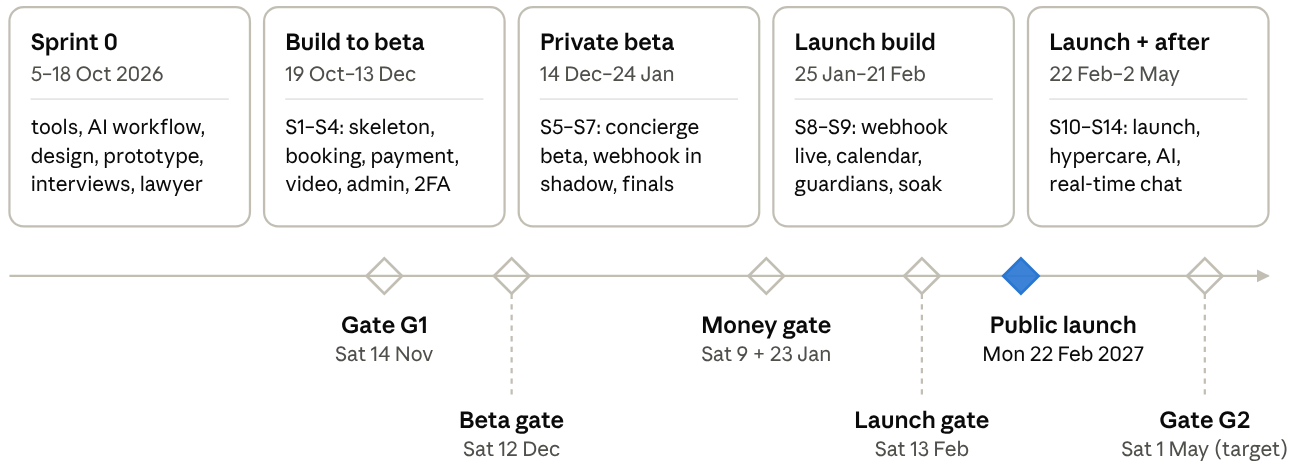
Your product-owner duties (capped at ~3 h a week)

The PO decides **what** gets built and in what order. The area owner decides **how**.

1. Keep one ranked backlog. Every item has a user story ("As a student, I want... so that...") and acceptance criteria written as Given / When / Then.
2. Set the sprint goal and agree each person's weekly deliverables at sprint planning.
3. Accept or reject finished work at the Saturday demo. **"Done" means accepted by you, not just merged.**
4. Answer **decision-needed** issues within 24 h on weekdays.
5. Guard scope: a new idea goes into the *next* sprint, or replaces work of equal size.
6. Run the gates (G1, beta, money gate, launch, G2) with the evidence R&D collects.
7. Talk to users: 3 student interviews before G1, 5 by the beta.
8. Approve spending, sign legal and payment documents, and run the weekly payout.
9. Protect the team's time: no messages expecting answers after 23:00 PKT or on declared exam days.
10. Post a 5-line update in Discord every Sunday: progress, metrics, risks, decisions, next week's goal.
11. Act as data steward: own the access matrix and decide who gets production access (you and the AI Engineer only).

4. Timeline

Beta opens 14 Dec, public launch 22 Feb 2027; each phase ends at a gate



timeline · 5 phases, 6 gates and milestones

Phases are not drawn to scale. A **calibration checkpoint on Sat 31 Oct** decides whether these dates hold or Plan B applies: beta Mon 8 Feb, launch Mon 5 Apr 2027, which still catches the German April-July application window. The fallback launch date, if the launch gate fails, is Mon 15 Mar 2027 (after Eid).

5. Week-by-week plan and weekly deadlines

Every week has two fixed deadlines: **Wednesday 22:00 PKT**, when each deliverable is marked on track, at risk or blocked, and **Saturday 11:00 PKT**, when finished work is demoed and accepted. "By Wed" means pushed by Wed 22:00; "by Sat" means ready to demo at the Saturday sync. Item codes (M1, L2...) refer to section 2. The day-by-day plan for weeks 1-6 is in the **Day-by-day plan** tab.

Sprint capacity (estimate)

A sprint is a fixed 2-week block. Exam and holiday weeks are planned at lower capacity. Everyone enters their real exam dates in the team calendar **by Wed 7 Oct**, and you adjust S2 and S7 at the S0 review.

Sprint	Dates	Capacity	Why	Coding h
S0	5–18 Oct	100% (set-up)	Accounts, design, interviews	~15 + 8 design
S1	19 Oct–1 Nov	100%	–	32
S2	2–15 Nov	~50%	Mid-terms (estimate, e.g. PIDE 2–10 Nov); Iqbal Day 9 Nov	16
S3	16–29 Nov	100%	–	32
S4	30 Nov–13 Dec	100%	–	32
S5	14–27 Dec	~60%	25–26 Dec holidays	19
S6	28 Dec–10 Jan	~80%	New Year; exam preparation	26
S7	11–24 Jan	~30%	Finals (estimate)	10
S8	25 Jan–7 Feb	100%	Semester break (estimate)	32
S9	8–21 Feb	~70%	Ramadan expected from ~8 Feb	22
S10	22 Feb–7 Mar	~70%	Launch + 2 weeks of close watching ("hypercare")	–

The weekly plan

Wk	Dates	Goal	Frontend + UI/UX	Backend (you)	AI Engineer	R&D Specialist	Testing focus
1	5–11 Oct	S0: team, tools, AI workflow	User-flow maps Tue ; 8 low-fi wireframes Sat ; landing page + waitlist Sun	Private repo + board Tue ; stack decision record Tue ; production project (waitlist only) Thu ; scaffold planned Thu , built Sat ; branch protection + CODEOWNERS Sat	AI config files merged Thu 20:00 ; runs AI workflow workshop Wed 7 Oct ; filter test list Thu ; fact-card template Sat	Interview script + screener Tue ; lawyer requests Thu ; mentor outreach list (50) Sat ; 2 interviews Sun	Definition of Done; CI required; "no real data in AI tools" agreed
2	12–18 Oct	S0: prototype, schema, legal route	High-fidelity R1 flows Tue ; app shell (M3) Wed ; clickable prototype Fri 16 Oct	Staging Mon ; schema + migrations Tue ; sign-in + roles Wed ; Safepay notes in own words Thu ; lawyer consult Sat	Filter tests + function Tue ; fact-card CI check Wed ; Sentry + test alert Sat	8 interviews total Sun ; legal question list Tue ; R1 participants booked Sat	Filter test list: 100% blocked, false positives tracked
3	19–25 Oct	S1: skeleton part 1	Usability R1 Mon–Wed ; Figma fixes Thu ; sign-up/login + 18+ gate Sat	Access matrix by hand Mon ; profile tables Tue ; RLS policies pass tests Wed ; private bucket (fake docs) Sat	Events table Mon ; RLS tests from matrix Tue ; break-it review Wed ; filter on bios Sat	R1 notes + synthesis Wed ; 2 mentor R1 sessions Thu ; first 3 video checks Sat	Usability R1; RLS break-it; clock change Sun 25 Oct
4	26 Oct–1 Nov	S1: skeleton done + calibration	Mentor application (M5) Wed ; directory + profile (M6) Sat	Approve/reject Mon ; gigs + RLS Tue ; search Wed ; SECP filing Fri ; calibration Sat 31 Oct	2FA (M2) Tue ; preview deploys Sat	3 more video checks Tue ; test-case sheet Thu ; matching form Sat ; runs test bash #1 Sun	Test bash #1: sign-up, apply, approve, directory
5	2–8 Nov	S2 (~50%): booking request	Booking request screens Thu ; wired Sat	Booking state machine + no double booking + 36-h rule Thu ; Safepay sandbox checkout Sat	Email set-up Mon ; time-zone tests Wed ; ledger tests (before code) Thu ; job worker Sat	Ready-to-pay form Mon ; interview synthesis Thu ; G1 draft Sat	Overlapping accept refused; time-zone matrix

Wk	Dates	Goal	Frontend + UI/UX	Backend (you)	AI Engineer	R&D Specialist	Testing focus
6	9–15 Nov	S2: payment v0 + G1	Checkout return pages Mon ; status badges Wed	Ledger + lock trigger Wed ; mark paid Thu ; chairs G1 Sat 14 Nov	Email templates Mon ; My bookings Wed ; filter on notes Thu	10 mentors checked Wed ; presents G1 evidence Sat	Sandbox checkout success/fail/cancel; ledger tests
7	16–22 Nov	S3: end-to-end session	Session page Wed ; payment-pending UI Sat	Refund + dispute states Wed ; reviews LiveKit PR Sat	LiveKit room Wed ; 72-h auto-confirm + emails Sat	Lawyer drafts Thu ; 12 mentors checked; beta cohort of 50 Sat	End-to-end on staging; video-failure path
8	23–29 Nov	S3: production hardened + alpha	Polish Wed ; legal pages Sat	Production hardening Wed ; deploy + alpha Sat 28 Nov ; Safepay production form Sat	In-app admin Thu ; reminder job Sat ; nightly backups Sat	Safepay documents Thu ; manual-step runbooks Sat	Role matrix, IDOR, private documents; dry run starts Sat 28 Nov
9	30 Nov–6 Dec	S4: beta-ready (part 1)	Usability R2 Tue–Thu ; fixes Mon 7 Dec	Real uploads open Mon 30 Nov ; alerts Wed ; live keys if approved Sat	k6 smoke Sat 5 Dec	Verifies uploads; Urdu templates Sat ; webinar #1 Sat 5 Dec	R2; game day #1 Sun 6 Dec
10	7–13 Dec	S4: beta gate	Fixes Mon ; invite screens Wed	Code freeze Wed 9 Dec 22:00 ; go/no-go Sat 12 Dec	Restore drill #1 Wed 9 Dec	16 mentors verified (≥ 2 female per corridor) Sat ; support rota	Regression Thu ; fix buffer Fri ; gate Sat
11	14–20 Dec	S5: PRIVATE BETA OPENS Mon 14 Dec	Top-5 beta fixes Sat	Mark paid twice daily; webhook signature + inbox in shadow mode by Sat 19 Dec	Shadow-compare job Sat	Support; post-session survey; webinar #2 (UK) Sat 19 Dec	Daily production smoke; signature test with a real Safepay payload
12	21–27 Dec	S5: shared payment function	Checklist page template Wed	Shared payment function deployed Wed 23 Dec ; first payout Tue 22 Dec	Payment-event job in shadow Wed	8 beta interviews Wed ; Germany PG fact cards	Webhook fault suite; no money deploys 24–28 Dec
13	28 Dec–3 Jan	S6: reconciliation + calendar	Calendar UI Sat	Reconciliation Sat 2 Jan	Calendar slots + instant booking Sat ; tests for L3–L5 written first, Sun 3 Jan	UCAS last-call campaign Mon ; 22 mentors; UK PG cards	Slot race; reconciliation

Wk	Dates	Goal	Frontend + UI/UX	Backend (you)	AI Engineer	R&D Specialist	Testing focus
14	4–10 Jan	S6: guardian + checklist pages	Checklist pages Sat 9 Jan	Hold expiry Wed ; guardian RLS Sat	Guardian API Sat	Checklist content final Wed 6 Jan ; R3 participants booked	Money gate review Sat 9 Jan
15	11–17 Jan	S7: maintenance (finals)	Bug fixes within 48 h	Payout Tue; fixes	Monitoring	Extra support on UCAS deadline Wed 13 Jan	Hotfixes only
16	18–24 Jan	S7: maintenance	Bug fixes	Guardian RLS frozen Sat 23 Jan	Shadow-mismatch report Sat	Fact-card re-check starts	Money gate final Sat 23 Jan
17	25–31 Jan	S8: webhook live Mon 25 Jan	Guardian UI Wed ; thread UI Sat	Switch live + watch Mon ; intro call + refund rules Wed 27 Jan	Attendance evidence Sat	Launch marketing plan Mon ; webinar #3 Sat 30 Jan	Game day #2 Sun 31 Jan
18	1–7 Feb	S8: freeze Sun 7 Feb	Intro-call UI Wed ; co-runs R3 Mon–Thu	Rate limits + secrets rotated Sat 6 Feb ; release tag Sun 7 Feb	k6 launch load + stress Sat	Usability R3 ; 30 mentors	Soak test starts Sun 7 Feb
19	8–14 Feb	S9: release candidate	R3 fixes Wed	Release checklist; go/no-go Sat 13 Feb	Soak ends Wed 10 ; restore drill #2 Wed	Launch posts scheduled Sat	Full regression Mon–Wed ; security re-check
20	15–21 Feb	S9: launch preparation	Launch page Wed	Runbook Sat ; deploy freeze Sat 20 ; LiveKit paid plan Mon 15	Alert tuning Wed	Mentor briefing Sat 20 ; launch emails	Production smoke Sun 21 Feb
21	22–28 Feb	S10: PUBLIC LAUNCH Mon 22 Feb	Fixes within 24 h	On call; daily reconciliation; payout Tue	Monitoring	Urdu support daily; launch webinars 27–28 Feb	Daily production smoke; incident log

Turning weekly deliverables into daily tasks (from week 7)

Day-level plans four months ahead would be fiction. Instead, at every sprint planning each person spends 15 minutes on these steps:

1. Copy your two weekly deliverables from the table above into GitHub issues.
2. Work out your **building budget**: committed hours × the sprint's capacity, minus fixed slots (sync, standups, test bash, reviews, the learning task).
3. **Plan only 80% of it**, or 70% in gate weeks 10, 18 and 19.

4. Split each deliverable into tasks of 1, 2, 3 or 4 hours. Tag each one **CC** (Claude Code), **AG** (Antigravity), **chat** or **hand**. Use at most 3 CC sessions a week.
5. Put each task on a day:
 - Anything someone else needs is due by Wednesday.
 - Money and security tests come before the code.
 - Fridays stay empty as catch-up days.
6. If the draft is over budget, fix it **at planning**: cut scope, swap owner or move a task. Don't find out on Friday night.

The Day-by-day plan tab shows a worked example.

6. How we work: daily rhythm and weekly meetings

The team meets live for only 2 hours a week. Everything else is async and written down, which suits four students with different timetables.

The weekly time pattern

Day	Normal week	Exam week	Why
Mon	2 h	1 h	One evening session; fits inside one Claude 5-hour window
Tue	2 h	Off	Exam weeks need full days off
Wed	1.5 h + checkpoint 22:00	1 h + checkpoint	Mid-week check: is anything at risk?
Thu	1.5 h	1 h	Evening session
Fri	Off (you: 1 h of PO work)	Off	Rest, Jumma, family; also the catch-up day
Sat	4 h: 11:00–12:00 sync , then a 3-h build block	2.5 h	The long block is for heavy Claude Code work
Sun	1.5 h: 20:00–21:00 test bash + reviews	1 h	Tests the week's merged work before Monday
Total	12.5 h (you: 14 h)	6.5 h	

Everyone can shift their own weekday times. The Wednesday and Saturday checkpoints never move.

Rituals

Ritual	When	Length	Rules
Async standup	Every working day, posted by 23:00 PKT	3 min	In Discord <code>#standup</code> , using the template below. Two missed days in a row get a friendly message from you
Mid-week check	Wed 22:00 PKT	2 min	Each deliverable marked on track / at risk / blocked. You decide on every at-risk item by Thu 22:00: cut scope, swap owner or move the date
Weekly sync	Sat 11:00–12:00 PKT (in Ramadan, after Taraweeh)	60 min	Google Meet. Free group calls stop at 60 minutes (Techzine), so keep to the agenda
Sprint review + retro + planning	Second Saturday of each sprint, 11:00–13:00	2 h	Review 45 min, retrospective 30 min, planning 45 min
Test bash	Sun 20:00–21:00 PKT	60 min	R&D runs it; everyone tests staging on their own phone. Two game days (Sun 6 Dec, Sun 31 Jan) replace it with deliberate fault drills
Bug triage	Monday, async	15 min	R&D + you set a severity (S1–S4) and an owner for each new bug
Pairing	Once a week	45 min	Rotating pairs; you pair with everyone once a sprint

Standup template:

Done: <what finished, with issue #>

Next: <exactly one task for the next working day, issue #, tool tag CC/AG/chat/hand>

Blocked: <none / what, and who can unblock>

Hours today: <n> AI: <Claude weekly % / "AI-out until ..." / "coursework heavy this week">

Weekly sync agenda (60 min):

Minutes	Item	Led by
0–5	Numbers: waitlist, verified mentors per corridor, sessions, revenue, open S1/S2 bugs	You
5–30	Demos: each person shows finished work and explains one thing in their own words; you accept or reject	Each owner
30–40	Quality: bugs, flaky tests, next big test event	R&D
40–50	AI usage: weekly Claude %, limits hit, Claude Code sessions planned for next week	QA-on-duty person
50–60	Risks, decisions needed, next week's deliverables confirmed	You

Task board (GitHub Projects, free)

Column	Rule
Backlog	Ranked by you only
Ready	Has acceptance criteria and an estimate
In progress	At most 2 items per person
In review	PR open; first review within 24 h
Test on staging	Checked at the next test bash
Done	Accepted by you at the Saturday demo

Custom fields:

- Sprint (2-week iteration)
- Estimate h (1, 2, 3 or 4 only)
- Actual h
- Due day
- Priority (Must/Should/Could)
- Area (FE/BE/AI/R&D/Ops)
- Item code
- Tool (CC/AG/chat/hand)

Labels:

- S1 – S4
- sensitive (money, auth, RLS, migrations, CI, admin)
- real-data (never handed to an AI tool)
- claude-heavy
- learning
- decision-needed
- blocked
- corridor-DE , corridor-UK

Estimates.

- Size every task in **hours with AI help**.
- Writing tests is part of the task, not a separate one.
- Log Actual hours when you close an item.
- At each sprint review, you compute the **estimate-accuracy ratio** (actual ÷ estimate).

Calibration on Sat 31 Oct. Use the ratio from Sprint 1:

- **1.2 or less:** keep the plan.
- **1.2-1.5:** keep the dates and cut 5 h of polish.
- **Above 1.5:** switch to **Plan B**: beta Mon 8 Feb, launch Mon 5 Apr 2027.

Communication tools (all free):

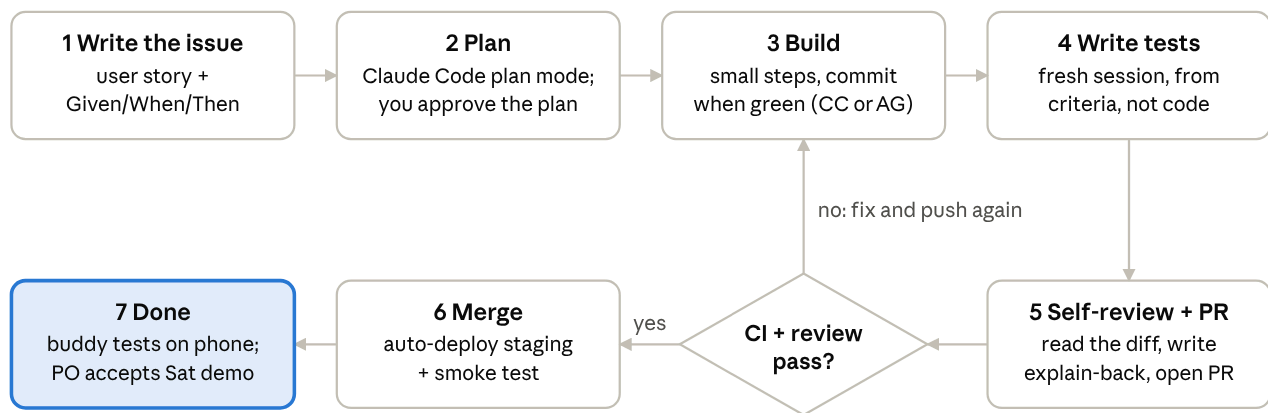
- **Discord:** #standup , #dev , #design , #research , #bugs , #alerts , #decisions , #support-log
- **GitHub:** code, issues and the board
- **Google Meet:** the weekly sync

- **Google Drive/Forms/Calendar:** coded interview notes and the team calendar
- **Figma Starter:** design
- **WhatsApp Business app** on a dedicated number: inbound customer support only

Quiet hours: nobody is expected to reply between 23:00 and 09:00 PKT, or on declared exam days.

7. Development workflow: from task to done

A change counts as done only after plan, build, tests, review and PO acceptance



development workflow · 7 steps, 1 decision

A typical task takes 2–3 hours. Each step:

- **Writing the issue:** 15–30 min.
- **Planning:** 15 min.
- **Building:** 60–120 min.
- **Tests:** 30–45 min, in a fresh session that sees only the public interface.
- **Self-review:** 20 min. Run `/code-review` and read the whole diff.
- **The pull request:** use the PR template, which has "How I tested", "AI use" and "Explain-back" sections.
- **Review:** money, auth and RLS changes need code-owner approval, and the reviewer does the break-it check.
- **Done:** a task is done only when you accept it at the Saturday demo, never just because it was merged.

8. Engineering practices

Use boring, managed technology. Every hour not spent running servers goes into the product.

Stack

Part	Choice	Why
Web	Next.js (App Router) + Tailwind + <code>next-intl</code> , on Vercel , functions in Frankfurt (<code>fra1</code>)	One codebase for web and phone (installable PWA); public pages served from a CDN near users
Database, sign-in, files	Supabase , Frankfurt: Postgres + pgvector, Auth (email, Google, 2FA), private Storage, Row Level Security	Saves an estimated 150–250 h of building auth, storage and chat yourselves. Data stays in the EU, which suits GDPR for mentors
Money, booking state, jobs, AI	One small FastAPI service on Render Starter (\$7/month, Frankfurt), with the job worker as a separate process	Anything involving money, another person's data, secrets or AI goes through FastAPI, never directly from the browser
Video	LiveKit Cloud : free Build plan in development; Ship (\$50/month) from Mon 15 Feb	The free allowance is a hard cap: when it runs out, new calls fail (LiveKit)
Payments	Safepay (cards 2.9% + Rs 30; Raast/wallets 1.5%)	Stripe doesn't support Pakistan and PayPal isn't available there
Email	Resend free tier (3,000/month, 100/day)	Supabase's built-in email is not for production
Errors + uptime	Sentry (free through the GitHub Student Pack)	One tool for errors, one uptime monitor and one cron monitor

Not used: Redis, Socket.IO, Alembic, SQLAlchemy models or a self-built chat server. Supabase migrations are the **only** source of truth for the database schema.

Repository layout (one monorepo)

```

apps/web/           Next.js: (public) pages, (app) dashboards, (admin) panel
apps/api/app/       FastAPI: routers/, domain/ (pure business rules), jobs/,
integrations/       (safepay, livekit, resend)
supabase/           migrations/ (schema, RLS, functions), seed.sql, tests/
(pgTAP + access_matrix.yaml)
packages/shared/    shared types and enums
ai/kb/cards/        fact cards (R&D edits them in GitHub's web editor)
tests/e2e/          Playwright      tests/load/    k6
docs/               adr/ (decision records), runbooks/, vendor/safepay.md,
testing/
scripts/            seed_demo.py (fake data only)

```

Environments

	Local	Staging	Production
Database	<code>supabase start</code> (Docker)	Supabase Free, Frankfurt	Supabase Pro (\$25/month), Frankfurt
API	<code>uv run uvicorn</code>	Render (Starter during test windows)	Render Starter + separate worker
Payments	Stub gateway	Safepay sandbox	Safepay live
Data	Fake (<code>seed_demo.py</code>)	Fake only, including fake documents	Real

Developer laptops link the Supabase CLI to **staging only**. Production migrations run only through the release workflow, which needs your approval.

GitHub set-up (verified)

- **Where the repo lives.** A private repo under your personal account with **GitHub Pro**, free through the [Student Developer Pack](#). The other three join as collaborators. Branch protection and code owners work on private repos with Pro ([GitHub docs](#)).
- **Rules on `main` :**
 - a pull request is required
 - 1 approval, plus code-owner review on code-owned paths
 - the latest push must be approved by someone else

- CI must pass
- **"Do not allow bypassing"**, so the rules also bind you as admin
- **CODEOWNERS = you + the AI Engineer.** This covers migrations, database tests, `apps/api/app/domain/`, `payment/webhook/admin` code, jobs, CI workflows and the AI config files. Your own sensitive PRs therefore need the AI Engineer ([GitHub docs](#)).
- A **sensitive-pr** CI check fails if sensitive paths change with fewer than **2 approvals from people other than the author**.
- **CI minutes.** GitHub Pro includes 3,000 Actions minutes a month for private repos ([GitHub](#)). Expected use is about 1,600. Run `pnpm test:fast` locally before pushing.

Branches, PRs and decisions

- **Branches.** Short-lived: `feat/<issue>--<slug>`, merged within 3 days. Conventional Commits (`feat:`, `fix:`, `test:`). Squash-merge.
- **PRs.** Under ~300 changed lines. First review within **24 h** on weekdays. On money and RLS PRs, the **reviewer** does the "break-it check": flip one condition and confirm a test fails.
- **Deploys.** Merging to `main` auto-deploys to staging. Production is a tagged release you approve.
- **Decisions.** A stack or data-model change needs a one-page decision record (ADR) in `docs/adr/`. Every decision gets one line in `docs/decisions.md`.

Code rules everyone follows

- **Money** is stored as **integer paisa**, never a decimal. The server computes every price; the browser never sends one.
- **Time** is stored in **UTC** and shown in each viewer's time zone. Domain functions take `now` as a parameter, so tests can jump to any moment, including the clock changes on 25 Oct 2026 and 28 Mar 2027.
- **Every table has RLS on, deny by default.** Money and booking tables have **no** user write policies; only FastAPI writes them.
- **The ledger is append-only.** A database trigger blocks edits and deletes, even by the table owner.
- **All user text** goes through translation files and is rendered as plain text (no HTML).
- **Validation** happens everywhere: zod on forms, Pydantic on API bodies, database `check` constraints as the last line.

Definition of Done (every task)

- ☐ Acceptance criteria met; every criterion has a test
- ☐ CI green: lint, types, tests, coverage gate, test-guard, sensitive-pr
- ☐ Reviewed (2 non-author approvals if **sensitive**); author can explain every changed line
- ☐ Test buddy PASS on a real phone; axe shows no serious accessibility issues
- ☐ New table: RLS on + rows added to the access matrix + tests pass
- ☐ Deployed to staging, smoke test passed
- ☐ Docs or decision record updated if behaviour changed; Actual hours logged
- ☐ **Accepted by the PO** at the Saturday demo

Data rules for the whole team

1. **No real user data in any AI tool:** Claude, Claude Code, Antigravity or any other. That covers names, contacts, grades, budgets, documents, chats, payment exports, support messages and un-anonymised interview notes.
2. AI may help with templates using placeholders, general questions, anonymised needs ("UK, postgraduate, CS, budget band B") and synthetic test data.
3. Interview notes use participant codes (P-S01...). The key to the codes lives in one sheet that only R&D and you can open.
4. Identity documents are seen by a human only: on a video call before Sat 28 Nov, in the private bucket after.
5. Nothing from production is downloaded to a laptop or phone. Payout sheets show only the last 4 digits of each IBAN.
6. Only you and the AI Engineer can log into the production Supabase dashboard. Its lowest role can still view and edit data, and a read-only role exists only on Team/Enterprise plans ([Supabase](#)). R&D uses the in-app admin page.

9. Claude vs Antigravity: which tool for which task

Claude Code does MAJOR work, at most 3 sessions per person per week. Antigravity does MINOR work. Claude chat drafts text. A human does secrets, payment settings, the access matrix, the webhook signature check and every merge. The full rules, prompt templates and ready-to-commit config files are in the **AI coding playbook** tab.

The limits you are planning around (verified)

- **Claude Pro** includes Claude Code. Chat and Claude Code **share** one usage limit, which resets every 5 hours. Pro gives roughly 45 chat messages **or** about 10–40 Claude Code prompts per 5 hours, and a weekly limit applies on top ([Claude Help Center](#), [Pro usage](#)).
- **Which model?** The same Help Center article says Pro users get Sonnet in Claude Code, not Opus. Check `/model` on your own account; the plan assumes Sonnet.
- **Google Antigravity** is free for individuals, with a usage quota that refreshes **weekly** ([Antigravity plans](#)). The exact numbers are not published.
- **Age:** both tools require users to be 18 or older.

The MAJOR vs MINOR rule

A task is **MAJOR** (Claude Code) if **any** of these is true:

1. It is a new feature, screen, endpoint, job or test suite.
2. It changes more than 2 files or about 100 lines.
3. It touches a **sensitive area**: payments, ledger, refunds, payouts, auth, 2FA, verification documents, the contact filter, admin, migrations or CI.
4. It changes a shared contract: the schema, an API shape or shared translation keys.
5. It is a bug you could not understand within 30 minutes.

A task is **MINOR** (Antigravity or by hand) only if **all** of these are true:

- it touches at most 2 files and about 50 lines
- it is not in a sensitive area
- you already know exactly what to change

Sensitive files are **never** edited in Antigravity. Never run both tools on the same branch at the same time.

Routing at a glance

Claude Code (MAJOR)	Antigravity (MINOR)	Claude chat	By hand
New screens, endpoints and jobs; booking and money logic (you approve every edit); RLS tests from the access matrix; test suites from acceptance criteria; k6 and fault-test scripts; fake seed data; hard bugs	Styling, spacing, copy and icon changes; one-file fixes; one extra test case or test row; lint fixes; a daily click-through of staging by the browser agent (a human watches the recording)	Acceptance criteria, ADR drafts, research briefs from official pages (every fact then checked by R&D), English/Urdu UI copy, test-case tables, summaries of anonymised notes	Secrets, Safepay and hosting dashboards, the access matrix, <code>verify_signature</code> , the break-it check, merges, usability sessions, Urdu checks by a native speaker

Five rules that make 4 Pro accounts enough

1. **Tag every task** CC / AG / chat / hand at sprint planning.
2. **At most 3 Claude Code sessions per coder per week** (2 for R&D). A session is at most 2 h on weekdays and 3 h on Saturday.
3. **Split heavy tasks:** a 20–30-minute plan-mode session on a weekday, the build on Saturday, and the tests in a fresh session.
4. **Sonnet by default.** Use `/clear` between tasks and check `/usage` at the start and end of every block. In gate weeks (10, 18, 19), keep 30% of your limit in reserve.
5. **Hit a limit?** Write a handoff note in the draft PR, post "AI-out until ..." in standup, and switch to MINOR or by-hand work.
 - A teammate with budget can take the MAJOR task on **their own** account. Never share logins.
 - If limits block the critical path two weeks running, you may approve paid usage credits with a monthly cap.

Safety rules (the short version)

- **Turn off** Claude's "help improve Claude" setting and Antigravity's telemetry. Each person posts a screenshot by Sun 11 Oct.
- **Claude Code:** plan mode is the project default, and `.env` files, `supabase db push`, `supabase db reset --linked`, `supabase link`, `vercel` and `gh secret` are blocked in the committed settings. Never use `bypassPermissions` or `--dangerously-skip-permissions`.
- **Antigravity** has had published security problems: in Nov 2025 researchers (PromptArmor) showed a prompt injection stealing `.env` keys through its browser agent, and a sandbox escape was fixed in Feb 2026. So:
 - set terminal commands to "Request Review" and keep the sandbox on
 - keep browser tools off except for staging tests, with only `localhost` and the staging domain allowed
 - open only your own repository
 - update the tool weekly
- **Treat untrusted text as data:** web pages, issues, PDFs and AI output can all contain hidden instructions. Never tell an agent to "follow the guide at <URL>".
- **Explain-back rule:** if you can't explain a line, it doesn't get merged. One task a week per person is done without AI, or with AI only as a tutor.

10. Testing strategy

Testing takes about 20% of everyone's time and grows in three kits: a minimum kit for the beta, a fuller kit for launch, and AI evaluation only once an AI feature exists. The four required test types each have fixed dates and pass rules. Test cases, scripts, the fault table and gate checklists are in the **Testing playbook** tab.

Daily and feature-wise testing

- **The author writes the tests in the same PR.** Every acceptance criterion has a test. For money, booking and RLS code, the tests come **first**, written in a fresh session that sees only the public interface.
- **CI blocks a PR** if coverage of the business-rules folder (`apps/api/app/domain/`) falls below **90% of lines / 85% of branches**. It also fails if someone skips or deletes a test without approval (test-guard).

- **A test buddy** (not the author) checks each feature on a real phone within 48 h: the key cases plus 10 minutes of "try to break it".
- **After every deploy**, an automatic 5-minute smoke test runs: pages load, logins work, search returns results, and an admin route without 2FA is refused.
- **Sunday test bash** (20:00–21:00): R&D gives each person a 40-minute focus, such as "booking across time zones".

The three kits

Area	Beta kit (by Sat 12 Dec)	Adds for launch (by Sat 13 Feb)
Unit	pytest + Hypothesis (fee maths, state machines, refunds, time zones); Vitest + React Testing Library	Tests for L1–L7, written before the code in S6
Database	pgTAP (tests that run inside Postgres): access matrix, RLS on every table, ledger, contact filter	Guardian and thread rules
End-to-end	Playwright + axe: 5 journeys	10 journeys
Performance	k6 smoke with a fake payment gateway; Lighthouse; a real phone	k6 load + stress; 72-h soak
Reliability	Sentry + 1 uptime monitor; restore drill #1; rollback practice	Restore drill #2; production rollback; money gate
Security	gitleaks, Dependabot, dependency audit, upload tests, IDOR checks (opening someone else's record by changing an ID)	One ZAP baseline scan; Schemathesis API fuzzing
Usability	Rounds R1 (prototype) and R2 (alpha)	Round R3

The four required test types

Type	What and when	Pass rule (team targets)
Usability	Moderated think-aloud sessions on Zoom or Meet, 40 min each. R1 19–22 Oct on the Figma prototype; R2 1–3 Dec on the alpha; R3 1–4 Feb on the release candidate. 6–7 people per round, including 2+ Urdu speakers, mentors in both countries and 2 parents in R3. 5 users find about 85% of problems (NN/g)	Critical tasks completed by 4 of 5 at R2. SUS score (a 10-question ease-of-use survey, average 68) $\geq$ 68 at R2, $\geq$ 75 at R3
Performance	Tested on "Slow 4G" with 4× CPU slowdown and a real mid-range Android: 89.77% of Pakistani mobile users are on Android (StatCounter). k6 smoke with 25 virtual users Sat 5 Dec ; launch load with 200 users + 20 bookings/min by Sat 6 Feb ; stress until it breaks	Core Web Vitals "good" (web.dev): LCP $\leq$ 2.5 s, INP $\leq$ 200 ms, CLS $\leq$ 0.1. Search p95 $\leq$ 500 ms (p95 = the time 95 of 100 requests beat). Money endpoints under 0.1% errors. Breaking point at least 2× the launch load
Reliability	Uptime target (SLO) 99.5% , which allows 3.6 h of downtime a month. Restore drills by Wed 9 Dec and Wed 10 Feb , then monthly. 72-hour soak test from Sun 7 Feb . Every scheduled job is checked for running on time, catching up after downtime and running only once	Drill restores real rows, a test login and one decrypted IBAN within 4 h. Soak: memory growth under 10%, 0 missed jobs, 0 stuck orders

Type	What and when	Pass rule (team targets)
Fault tolerance	22 deliberate failure scenarios, using staging-only fault switches and a fake gateway: webhook delayed, duplicated, lost, out of order or forged; gateway down; video drops; database down for 60 s; email provider fails; job runs twice; payout job crashes halfway; double booking; upload fails. Two game days (Sun 6 Dec, Sun 31 Jan): you break 4 things without warning, and the on-duty person must spot each from alerts	Money and access always fail closed (refuse rather than guess). No double ledger entries, no lost payments, no automatic refunds from missing data, recovery within the time stated in each scenario

Two ideas worth learning now:

- **Idempotency:** doing something twice has the same effect as doing it once, like pressing a lift button five times. Safepay retries webhooks, and if one event fails, *all* events go into its retry queue ([Safepay](#)). So the webhook stores each event once, **always answers 200** (401 only for a bad signature), and a job applies it.
- **Retry with backoff:** wait 1 s, then 2 s, then 4 s, and so on, plus a little randomness, and give up after a limit. Never blindly retry a payment: ask the gateway for the status instead.

Quality gates

Gate	Date	Must hold (highlights)
Pre-beta	Sat 12 Dec	0 open S1 bugs; payment and ledger tests green; 10-order dry run with 0 errors; restore drill passed; RLS tests green; admin 2FA on; R2 critical tasks $\geq 4/5$
Money gate	Sat 9 Jan + Sat 23 Jan	Webhook fault tests green; at least 20 orders with 0 mismatches in shadow mode → webhook goes live Mon 25 Jan
Pre-launch	Sat 13 Feb	Everything above + launch load passed + 72-h soak + restore drill #2 + ZAP scan with no High findings + SUS ≥ 75

Gate	Date	Must hold (highlights)
AI feature	Before any AI ships (after launch)	100-question golden set: refuses individual visa advice 100% of the time, correct $\geq 90\%$, cites an official source $\geq 95\%$, 0 prompt-injection failures

Bug severity

Severity	Meaning	Response
S1	Money wrong, personal data exposed, security hole, site down. If money or personal data <i>might</i> be involved, it is S1 until proven otherwise	Acknowledge within 2 waking hours; mitigate (rollback or kill switch) within 24 h
S2	Core flow broken for some users	Within 3 days
S3	Workaround exists	Next sprint
S4	Cosmetic	Backlog

11. Security, privacy and safety implementation

The rule behind every item below: money and access **fail closed**, documents are seen only by a human, and no personal data leaves your systems.

Topic	How it is built	When
Sign-in	Supabase Auth: email + password and Google sign-in; CAPTCHA (Cloudflare Turnstile) on sign-up	W2-W3
Roles	Student, mentor, admin (ops level for R&D, money level for you + AI Engineer); guardian at launch. Only the server can change roles	W2

Topic	How it is built	When
Row Level Security	On for every table, deny by default. You write the access matrix by hand; the AI Engineer generates tests from it in a separate session; the reviewer loosens one policy to prove a test fails. A second test logs in as every role and calls every table and endpoint. A new table without a matrix row fails CI	W3 onward
Admin 2FA	Authenticator-app codes enforced on admin pages and FastAPI admin routes (Supabase MFA)	W4
Identity documents	Until Sat 28 Nov: "show, don't send" video check, metadata only. From Mon 30 Nov: private bucket; a mentor can upload but never read back; admins get 2-minute signed links; every view is logged; files are deleted 30 days after the decision	W8–W9
Uploads	PDF/JPG/PNG only, at most 10 MB, file type checked from its content (not the extension), photo location data removed	W8
Bank details	IBAN encrypted in FastAPI; only the last 4 digits stored in clear text; payout sheets never show full IBANs	W8

Topic	How it is built	When
Payments	The server computes every amount. "Mark paid" needs a unique Safepay reference + the exact amount. The ledger is append-only. The webhook signature (HMAC-SHA512, header <code>X-SFPY-SIGNATURE</code>) is written by hand from Safepay's docs and tested against a real signed sandbox payload. Live and sandbox keys are kept separate	W6, W11
Contact filter	One database function with normalisation (Urdu digits, "O" for zero, spaces, zero-width characters). Hard blocks only for clear phone, email and messenger-link patterns; long digit runs get a soft flag for review, so prices and dates still work. Strikes only after a human confirms. AI moderation is added after launch	W2-W6
Under-18s	Not allowed in the beta (18+ gate). At launch: parent-linked accounts; the parent approves and pays; read-only parent view; guardian rules frozen and tested by Sat 23 Jan	Launch
Rate limits	Vercel firewall rule per IP + per-user limits in Postgres (bookings, uploads, messages); final values set before the release tag	W18
Secrets	Each platform's secret store + one shared password manager. Never in code, prompts or screenshots. gitleaks on every PR. Rotated before launch	Ongoing

Topic	How it is built	When
Backups	Supabase Pro daily backups (7 days) + a nightly encrypted dump of roles, schema, data and auth users to Cloudflare R2. The default dump contains no data (Supabase CLI), so the flags matter	W8
Privacy (GDPR-level)	A clear privacy notice; collect only what matching needs; data export and deletion on request within 30 days; a record of processing in docs/ ; sign Supabase's data processing agreement; breach runbook (72-hour notice under GDPR Art. 33)	Before real users
Female mentor option	Optional, self-declared; a "female mentors only" filter; never required, never shown otherwise	M6
Legal basics	Company or sole proprietor, ToS, privacy policy, refund policy, mentor contractor agreement, "admission never guaranteed", "no visa advice". No real money before these are published and the lawyer has approved holding money per order	Lawyer drafts Thu 19 Nov; published Sat 28 Nov

What mentors may not sell. Individual immigration advice for money is a regulated profession in the UK, and Germany restricts paid legal help (see the concept document). Mentors may help with university choice, statements of purpose, admission checklists and life abroad. They may **not** judge a buyer's visa chances, fill in visa forms or plan a refusal appeal. The mentor agreement and the AI rules both say so.

12. Beta and launch: gates and success metrics

Each gate is a go / narrow / stop decision you make on a fixed Saturday, using evidence R&D collects. The targets below are the team's own estimates; adjust them at G1 once real numbers exist.

Gate G1: Sat 14 Nov 2026 (are mentors and students real?)

Check	Target
Mentors ready	10 or more, at least 4 per corridor : right to earn checked by video, beta agreement signed
Students interviewed	25 or more, about half per corridor
Willing to pay	15 or more students who picked a real gig at its real price, at least 5 per corridor

Possible outcomes:

- **GO** on both corridors.
- **One corridor first**: if a corridor has fewer than 4 ready mentors, open the beta with the other one while recruiting continues.
- **STOP and rethink**: if there are fewer than 8 mentors or fewer than 8 ready-to-pay students, building pauses for one sprint.

Private beta go/no-go: Sat 12 Dec 2026

- All Must items M1-M15 accepted.
- 0 open S1 bugs, and at most 3 S2 bugs with workarounds.
- Payment and ledger tests green; the 10-order dry run had 0 errors.
- Restore drill #1 passed.
- Documents only in the private bucket; admin 2FA on; production dashboard limited to you + AI Engineer.
- At least 12 verified mentors (5+ per corridor, 2+ female per corridor). Support rota staffed.
- **Real money only if** Safepay live keys are active **and** the terms are published **and** the lawyer has approved. Otherwise the beta runs free or vouchered sessions.

The beta itself: about 16 mentors and 50 invited students (18+), Mon 14 Dec to Mon 22 Feb.

Public launch go/no-go: Sat 13 Feb 2027

- **Webhook**: live since 25 Jan with 0 reconciliation mismatches.
- **Launch items**: L2-L6 done, or deferred by you (L7 may be dropped).

- **Tests:** load test, 72-h soak and restore drill #2 passed. Secrets rotated and rate limits set before the release tag.
- **Mentors:** at least 25 verified, 10+ per corridor, 3+ per corridor who mentor in Urdu.
- **Beta evidence:** at least 30 paid sessions, average rating 4.5+, refund rate under 10%.
- **Fallback launch: Mon 15 Mar 2027**, after Eid, if this gate fails.

Launch checklist (week 20)

- ☐ Production smoke test passed Sun 21 Feb; deploy freeze from Sat 20 Feb
- ☐ LiveKit paid plan active from Mon 15 Feb (the free cap would block calls)
- ☐ Launch-day runbook: who watches Sentry, reconciliation, support and payouts each day
- ☐ Mentor briefing done Sat 20 Feb; waitlist email scheduled for Mon 22 Feb 09:00
- ☐ Rollback practised on production with a harmless change
- ☐ Status page ready; kill switches tested (pause new bookings; "mark paid" fallback)
- ☐ Launch webinars Sat 27 Feb (DE) and Sun 28 Feb (UK)
- ☐ Two weeks of close watching: S1 acknowledged within 1 h, daily smoke test + reconciliation

Metrics from day one

The metrics come from a first-party `events` table: no cookies and no consent banner. You review them every Saturday.

Metric	Why it matters
Funnel: waitlist → sign-up → booking request → paid → confirmed → rated	Shows where students drop off
Verified mentors per corridor, and mentor reply time	The supply side of the cold start
Paid sessions per week, repeat bookings, refund rate	Revenue and trust
Average rating and SUS score	Quality
Reconciliation mismatches, S1/S2 bugs, error budget used	Reliability

Metric	Why it matters
Claude usage %, actual vs estimated hours	Whether the plan is still realistic

Waitlist targets: 150 by G1, 400 by the beta, 1,000 by launch.

Gate G2 (target Sat 1 May 2027): 100 paid sessions with an average rating of 4.5+. If reached, plan Phase 2 (communities, AI finder, Arrival Kit). If not, fix conversion and mentor supply first.

13. Monthly budget

Development months cost about **PKR 22,600–40,000** (about \$81–143), mostly the four Claude Pro accounts. The beta/launch month costs about **PKR 51,200–58,800** (about \$183–210), plus Safepay fees deducted from collections. Prices were checked on official pages on 2 Oct 2026. Rate: PKR 280 per US\$ (actual 277.24 on 1 Oct, [Trading Economics](#)). Pakistani cards may add FX charges, so keep a 5–10% buffer.

Item	Development months	Beta / launch month	Source
Claude Pro × 4	\$80	\$80	\$20/month each (claude.com/pricing)
Google Antigravity	\$0	\$0	Free individual plan
Domain (.com)	~\$0.87	~\$0.87	\$10.46/year at Cloudflare
Web (Vercel)	\$0–20	\$20+	Hobby is non-commercial only; Pro \$20 per seat (Vercel). Seat count to confirm
Database (Supabase)	\$0, or \$25 if production starts in October	\$25	Pro: daily backups for 7 days (Supabase)
API (Render Starter)	\$0–7	\$7 (+\$7 staging in gate weeks)	Render

Item	Development months	Beta / launch month	Source
Video (LiveKit)	\$0 (Build, 5,000-minute hard cap)	\$50 (Ship), from Mon 15 Feb	LiveKit . Plan B: Daily.co, about \$19 at beta volume
Email (Resend)	\$0	\$0–20	Free: 100/day
Errors + uptime (Sentry)	\$0	\$0	Student Pack
LLM	\$0–10 (tests only)	\$0 until AI ships in March	Chosen in a model bake-off in S10
WhatsApp Cloud API	\$0	\$0 (after launch)	Inbound support only before then
Usability-test gifts	~PKR 1,000 per participant (estimate)	–	About 20 participants in total
Company registration and lawyer	One-off (quote in week 2)	–	SECP fee is payable within 7 days of filing (SECP)
Monthly total	≈ \$81–143 (PKR 22,600–40,000)	≈ \$183–210 (PKR 51,200–58,800)	Sum of the rows above. Once AI features and WhatsApp reminders ship (March), add roughly \$25–60 a month (architect's estimate)

Gateway fees come out of collections, not your card: about PKR 19,600 for 150 sessions at PKR 5,250 (half by card at 2.9% + Rs 30, half by wallet/Raast at 1.5%).

Does a busy month cover its costs? 150 sessions × PKR 5,000 × 20% = **PKR 150,000** platform revenue. Minus gateway fees and running costs, that leaves roughly **PKR 71,600–79,200**, before refunds, free sessions and tax. That's an estimate, not a forecast.

Ways to save:

- Claim the GitHub Student Pack in week 1 (GitHub Pro, Sentry, 1Password, BrowserStack).
- Annual Claude Pro billing works out at \$17/month, but needs \$800 paid up front.

- Switch LiveKit to Daily.co if money is tight.

14. Risks to the plan and fallback actions

The two biggest risks are optimistic estimates and paperwork delays (Safepay, lawyer, company registration), not code. Each risk below has an early warning sign, so you act before it bites.

#	Risk	Early warning sign	Fallback	Owner
1	Estimates too optimistic	Estimate-accuracy ratio above 1.2 at the Sat 31 Oct calibration	1.2–1.5: cut 5 h of polish. Above 1.5: Plan B (beta Mon 8 Feb, launch Mon 5 Apr 2027)	You
2	Exams and coursework	Someone logs under 60% of planned hours; exam dates missing after Wed 7 Oct	Backup owner takes the deliverable; S7 is maintenance only; Should items drop from the bottom	You
3	A member drops out or goes quiet	2 missed standups in a row; no commits for 7 days	Backup takes over; recruit a replacement through a university society (~1 week to onboard). If you are out, the AI Engineer acts as PO for up to 2 weeks (no money or legal decisions)	You / AI Engineer
4	Claude Pro limits (including coursework use)	"AI-out" twice in a sprint; weekly usage above 60% by mid-week	Teammate with budget takes the task on their own account; work by hand; paid credits with a monthly cap if the critical path is blocked two weeks running	AI Engineer

#	Risk	Early warning sign	Fallback	Owner
5	Scope creep	Issues added mid-sprint without removing equal hours	New ideas go into the next sprint or swap for equal hours; freeze scope at the Wednesday check	You
6	Safepay approval delayed	No reply 5 business days after submitting; no site visit booked by Fri 4 Dec	Beta runs free or vouchered sessions (never bank or Raast transfers); the webhook switch moves to the launch gate; no live payments by Sat 13 Feb → launch Mon 15 Mar	R&D / You
7	Legal or registration delay	No lawyer booked by Fri 16 Oct; drafts not received by Thu 19 Nov; no SECP certificate by Sat 21 Nov	Apply to Safepay as a sole proprietor; never take money without published terms	R&D
8	Data incident	A document arrives by email or WhatsApp; personal data appears in an AI prompt or a downloaded file	Delete the copy at once, log it, tell the person affected, fix the runbook at the next sync	You
9	Not enough eligible mentors	Fewer than 4 ready per corridor at G1; the UK pipeline is mostly Student-visa holders	Open one corridor first; widen to British/settled Pakistani-origin graduates and German permanent residents	R&D
10	Ramadan and Eid around launch	Team capacity below 60% in week 19; gate items still open Wed 10 Feb	Fallback launch Mon 15 Mar 2027, after Eid (~9 Mar)	You

Work that isn't code (owners and dates)

Workstream	Owner	Key dates
Lawyer + business entity	R&D (you sign)	Requests Thu 8 Oct; consult Sat 17 Oct; SECP name reservation Sun 18 Oct; filing Fri 30 Oct; tax number (NTN) + bank certificate by Sat 21 Nov
Terms, privacy, refund policy, mentor agreement	R&D (you sign)	Outline Tue 13 Oct; drafts Thu 19 Nov; published Sat 28 Nov; signed by beta mentors by Sat 12 Dec
Safepay merchant account	R&D paperwork, you sign	Sandbox Thu 15 Oct (it works before approval); documents Thu 26 Nov (Safepay document list); production form Sat 28 Nov; target live keys Fri 11 Dec
Mentor recruitment	R&D (you take screening calls if R&D is over 13 h/week)	50 contacts Sat 10 Oct, then ~25 a week. Targets: 4+4 by G1 → 8+8 by the beta → 15+15 by launch. Funnel estimate: 100 contacts → 10 verified
Student interviews	R&D (everyone helps)	12 by Sun 18 Oct; 26 by Sun 1 Nov; 8 beta interviews in W12
Fact cards (Germany + UK)	R&D	Template Sat 10 Oct; 10 Germany PG cards by Sat 21 Nov; all four checklists final Wed 6 Jan; re-verified monthly (older than 90 days = flagged)

Workstream	Owner	Key dates
Community seeding (cold start)	R&D + Frontend	Instagram, LinkedIn and a WhatsApp Channel Sat 17 Oct; mentors answer 2 public questions a week in student groups from Mon 2 Nov; free webinars Sat 5 Dec, Sat 19 Dec, Sat 30 Jan
Urdu customer support	R&D rota	Reply templates Sat 5 Dec; rota from Mon 14 Dec; reply within 12 h in the beta

15. Sources

The planning agents opened these pages on 2–3 October 2026. Hour figures, capacity factors and targets marked "estimate" or "team target" are our own planning numbers, not published facts.

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- [University of Bath: key dates for 2027 entry](#)
- [GOV.UK: Student visa, apply](#)
- [University of Passau: uni-assist deadlines](#)
- [PIDE academic calendar 2026](#)
- [GOV.UK: clock changes](#)

AI tools

- [Claude Help Center: Claude Code with Pro or Max](#)
- [Claude Help Center: Pro plan usage](#)
- [Claude pricing](#)
- [Google Antigravity: plans](#)
- [Antigravity browser docs](#)
- [METR: AI and experienced developers \(2025\)](#)
- [Peng et al.: Copilot experiment](#)

Stack, hosting and prices

- [Supabase pricing](#)
- [Supabase access control](#)
- [Supabase MFA](#)
- [Supabase CLI: db dump](#)
- [Vercel pricing](#)
- [Render pricing](#)
- [LiveKit pricing](#)
- [LiveKit quotas](#)
- [Trading Economics: PKR rate](#)

Payments, legal and registration

- [Safepay webhooks overview](#)
- [Safepay: verify HMAC signatures](#)
- [Safepay: documents required](#)
- [SECP: online incorporation](#)

GitHub and team tools

- [GitHub: protected branches](#)
- [GitHub: code owners](#)
- [GitHub Actions billing](#)
- [GitHub Student Developer Pack](#)
- [Techzine: Google Meet 60-minute limit](#)

Testing

- [NN/g: test with 5 users](#)
- [web.dev: Core Web Vitals](#)
- [StatCounter: mobile OS share, Pakistan](#)
- [Google SRE Workbook: error budgets](#)
- [MeasuringU: SUS](#)
- [k6 thresholds](#)
- [Supabase testing guide \(pgTAP\)](#)
- [OWASP Top 10:2025](#)